Smart Control of Traffic Light System Using Artificial Intelligence

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***Abstract*— Our lives have been made easier by the swift advancement of scientific advances across all domains, yet the road traffic accidents have increased over time and claimed countless lives. The average commute time in India's main cites has increased dramatically during rush hour. The typical traffic signal runs on a constant stream of instructions that are kept in memory. For over ten years, traffic control has been a challenge, and it will likely remain so for some time to come. In this system vehicle density at an intersection is measured and a signal switching algorithm is used to measure the proper time for the duration of the green and red lights, as well as the length of the vehicles waiting in each route at the intersection and also the next intersection to ensure a smooth flow of traffic.**

***Index Terms*—Artificial Intelligence,Smart Intersection,Road Traffic Control**

I. INTRODUCTION

The existing traffic control system that is in operation causes traffic congestion on several road networks. The influence of the traffic backup causes drivers to drive more randomly, increasing their risk of breaking the law or causing unintentional accidents. Additionally, unneeded fuel usage at traffic intersections contributes to an increase in air pollution. As a component of the air pollution issue, traffic congestion has a direct and indirect impact on the environment and the economy. Fuel usage and waiting times for transportation are also lost. The traffic signal timer we use now is the primary factor affecting the traffic system. The traffic density in real time is not taken into account by such traffic signal timings, which function based on static timers that are fixed to them. The traffic control system continues in the same phase and for the same amount of time. The existing traffic signal management system is not adaptable and has a set waiting time that varies based on the type of traffic at different times. Several vehicles can be efficiently controlled during rush hour, which includes the morning, midday, and evening hours.

The research offers a novel solution: artificial intelligence (AI)- driven traffic lights. These lights use sophisticated algorithms to assess traffic density and modify traffic signals appropriately. This technology has the potential to improve traffic flow, reduce congestion, and raise general road safety. For security purposes, a CCTV surveillance camera is positioned at the intersection of traffic.In order to maximise traffic flow and reduce traffic congestion, the number of vehicles and vehicle classification can be determined by taking a picture at the junction. This image can then be used to operate the traffic light based on the current traffic density in the area.

However, the core aim which was set to conduct this project is orderly introducing a novel methodology using artificial intelligence, which permit solving the problems of a traffic light system. It must be reached through the next objectives:

1) Traffic flow optimisation
2) These systems can adapt to changes in traffic patterns through the day
3) Pedestrian safety
4) It reduces idling time of vehicles at the intersection

II. LITERATURE REVIEW

1.Intelligent Traffic Light System Using Image Processing: A literature survey(2021 Conference)

-Finding methods for identifying cars from video cameras

-Creating an algorithm to count the number of vehicles on the road via image processing.

-Real-time analysis of traffic density is done by counting the number of vehicles on the road.

-Camera is fixed to a high angle at various crossings to take images of the traffic on the road and transmit the image signal to the image processing server at the central control center.

-Building a traffic light system is one way to address the problem of traffic congestion. This is due to the fact that it comes under the umbrella of intelligent traffic lights.

2. Artificial Intelligence Based Framework For Effective Performance of Traffic Light Control :

A Literature survey(A Conference 2021):- This model react in real time based on quantity of vehicles. This model used in locations with traffic lights (a two-, three-, or four-lane traffic light zone, for example). HD cameras and a real-time video processing device make up the system.

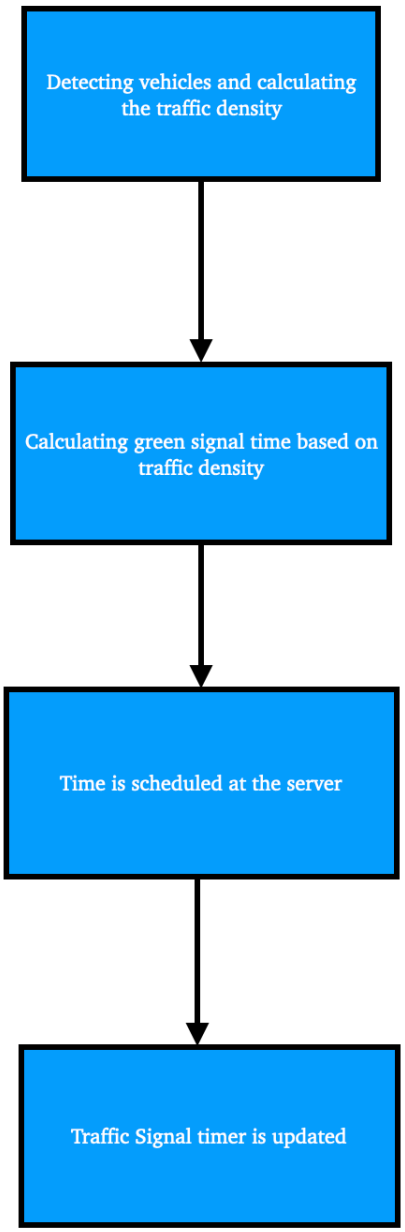
-Using YOLO2 this gives the total number of vehicles in each lane.
-This count number is processed once more in order to regulate traffic lights.
-The traffic light decision-making model, which regulates the traffic light depending on the overall number of vehicles that are in excess of the allotted space in that specific lane.

3.AI Driven Traffic Lights A Literature Survey(A conference 2023):

-In this model cameras are used capture the images on each road and sends to the computer.

-Computer/Raspberry Pi: This device is utilised to do preprocessing .
-LED Driver: An LED driver is a device that powers the LEDs that show the green and red signals

III. SYSTEM ARCHITECTURE



Caption

IV. PROPOSED SYSTEM

A.Calculating Vehicle Density

The function set_Time() is accountable for modifying the signal timings according to the quantity and kind of cars identified at the subsequent signal.To count the number of each kind of vehicle (cars, buses, trucks, rickshaws, and bikes), it first initialises variables. At first, these numbers are set to 0.The vehicles in the next signal (the one that will turn green next)are then cycled through. It verifies the kind of each vehicle and increases the associated count variable for each one.Following the completion of the vehicle count, the total time required for the green signal at the subsequent signal is computed by taking into account the counts of various vehicle categories. The average time for each kind of vehicle to pass the intersection is taken into account in this computation.The green signal of the subsequent signal in the sequence is then given the computed green time.

For debugging purposes, the function additionally prints the computed green time.This procedure is repeated for every signal change cycle.As a result, cars are detected at each signal indirectly by analyzing the cars that are waiting for the next signal before beginning to move. The number and kinds of these cars have an impact on when the next light turns green.

B.Signal Switching Algorithm

There are several traffic signals at this intersection, each of which directs traffic in a different direction.

To control the flow of cars, signals alternate between the green, yellow, and red states.

While the other signals stay red, the present green signal permits traffic to travel in its direction.

The green signal changes to yellow briefly to notify an upcoming signal change as its timer goes off.

Following the yellow phase, the subsequent light in the sequence turns green, enabling the flow of traffic in that direction.

The next signal turns green and the preceding green signal turns red, repeating the cycle.

Predefined parameters like default_Red, default_Yellow, and default_Green regulate the timing of signal changes as well as the length of the green, yellow, and red phases.The repeat() function controls the series of signal changes, modifying the 8ming of the signals and directing traffic flow accordingly. When creating the algorithm, the following factors were taken into account:

1)The time required for processing of the algorithm to calculate the amount of traffic and the length of the green light determine when the picture needs to be taken.

2)The number of lanes.

3)The overall count of vehicles in every category, comprising trucks, cars, motorbikes, and so on.

4)Traffic density computed with the above mentioned variables.

5)Additional time added due to the delay each vehicle encounters during startup and the non-linear rise in lag experienced by the vehicles in the back.

6) The average speed of each category of vehicle at the start of the green light.

The timing of when signal changes occur and how long signal changes happen is managed by predefined parameters such as default_Red, default_Yellow, and default_Green as well as the length of the green, yellow, and red phases. The series of signal changes is controlled by the repeat() function, which changes the timing of each signal and controls which signal changes go first or last. While creating the algorithm, the following factors were calculated: the computation time of the algorithm to calculate the number of cars and the time of the green light is the time when the picture needs to be taken, the number of the lane and the general number of each class of cars, such as lorries, cars, motorcycles, people: Traffic density calculated with the above variables, and the additional time on the rise of lag's non-linear exponential dependence given above.The signals alternate cyclically rather than in the direction that is densest first. This is compatible with the current system, which doesn't force users to alter their routes or cause confusion because the signals follow a preset schedule when they turn green one after the other. The signal sequence remains the same as it does in the current system, and care has also been given to the yellow signals.

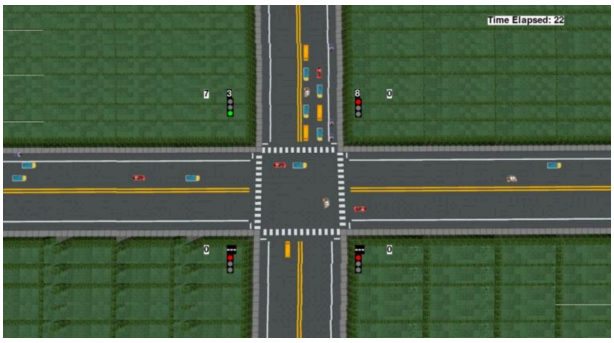
Signal sequence: Red—>Green—>Yellow—Red.

C. Simulation

Pygame was used to create a simulation from scratch that mimics real-world traffic. It helps to make the system easier to see and contrasting it with the static structure that is in place. There is a four-way stop with four traffic signals there.Each signal has a timer on top of it that shows how long it will take for a signal to change from green to yellow, yellow to red, or red to green. Next to each light is the number of cars that have crossed the crossing.

Every way is filled with cars, motorbikes, buses, trucks, and rickshaws. To make the simulation more realistic, some of the vehicles in the far right lane turn to cross the intersection.To make the simulation more realistic, some of the vehicles in the far right lane turn to cross the intersection. There are random numbers used when the car is constructed. Additionally, a

timer on it displays the amount of time that has passed since the simulation started.A screenshot of the simulation's ultimate result is displayed in below figure.



A cross-platform collection of Python modules called Pygame is intended for use in game development. It comes with sound and graphic libraries made specifically to work with the the programming language Python.Pygame enhances the great SDL library with capabilities. This enables users to write multimedia applications and games using all the features of the Python language. Pygame is incredibly versatile and compatible with almost all platforms and operating systems. It is available for free under the LGPL.

V. RESULTS

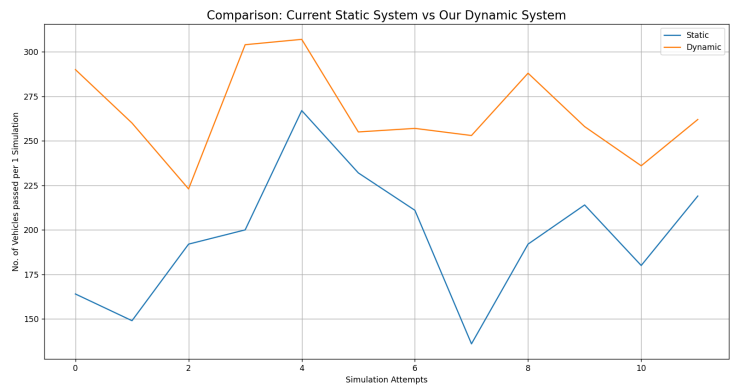
In this study we compare the number of vehicles passing through the intersection in current system and the implemented system considering random number of vehicles in each simulation.A total of 12 simulations were carried out each lasting for 5 minutes.

From the table given below it is clear that our implemented system could pass 3193 vehicles in an hour compared to 2356 vehicles in the static system, which is an increase of 837 vehicles.This translates to an increase in performance by 35% of the AI enabled traffic light system. On an average 70 more vehicles can pass in every simulation,reducing the waiting 8me of the vehicles and increasing the efficiency.

Current Static System					
Simulation No.	Lane 1	Lane 2	Lane 3	Lane 4	Total
1	59	77	17	11	164
2	20	48	31	50	149
3	77	85	23	7	192
4	78	73	22	26	200
5	107	82	38	40	267
6	81	104	31	16	232
7	77	85	27	22	211
8	19	53	25	38	136
9	60	43	57	32	192
10	35	68	70	41	214
11	52	30	23	75	180
12	73	80	19	47	219
					2356

Our Dynamic System					
Simulation No.	Lane 1	Lane 2	Lane 3	Lane 4	Total
1	118	122	23	27	290
2	121	101	20	18	260
3	80	98	30	15	223
4	115	150	19	20	304
5	118	132	28	29	307
6	100	92	33	30	255
7	134	73	18	32	257
8	111	83	33	26	253
9	118	106	27	37	288
10	92	87	56	23	258
11	81	72	43	40	236
12	79	124	39	20	262
					3193

Here is a graphical representation of the same.



It can be implemented economically and effectively.

VI. CONCLUSION

We are able to cut down on the vehicles' waiting times by using the dynamic model.
When compared to the current system, the performance of the proposed traffic control system using AI is improved by 35%. Using the AI model increases the functionality and reduces the time that cars have to wait at their next crossing.

At the traffic light itself, edge computing can be used to implement both of these.

It can be implemented economically and effectively.

VII. REFERENCES

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